

SOURAV SUMAN

Software Engineer | Backend & Full-Stack Developer

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Professional Summary

Final-year Engineering student with strong DSA foundations (200+ LeetCode problems solved) and a track record of independently designing, building, and debugging backend services — fraud detection APIs, WebSocket-based real-time systems, NLP pipelines, and CI/CD-driven deployments — all developed and version-controlled as public open-source repositories on GitHub. Comfortable working across Python, Java, and JavaScript, with a genuine curiosity for how systems work internally.

Education

B.Tech, Electronics & Communication Engineering

Bharati Vidyapeeth (Deemed University), Pune

2023 – 2027

CGPA: **9.1/10.0**

B.Sc, Data Science & Applications

Indian Institute of Technology (IIT) Madras

2024 – Present

CGPA: **7.2/10.0**

Technical Skills

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js, FastAPI, Flask, REST APIs
- **Database:** PostgreSQL, MySQL, MongoDB, Redis
- **System Design & Architecture:** JWT Authentication, WebSockets, Caching, Rate Limiting, Distributed Systems basics
- **DevOps & Tools:** Docker, GitHub Actions, CI/CD, Git, AWS (S3, EC2), GCP, Vercel, Render
- **Applied AI/ML (secondary):** PyTorch, Scikit-learn, Hugging Face, RAG Pipelines, LangChain, LangGraph

Certifications

Programming in Java — NPTEL (Certificate)

Data Analytics with Python — NPTEL (Certificate)

Projects

ROADSoS — Real-Time Emergency Response Platform

[GitHub](#) | [Live](#)

React 19 · FastAPI · WebSockets · Redis · Docker · Service Workers

- Designed a real-time **WebSocket/Redis**-backed location tracking system with trauma-center prioritization, achieving **sub-second** discovery latency.
- Debugged and fixed a critical **Redis race condition** using an atomic **Lua script**, eliminating data inconsistency under concurrent load.

FraudShield — Self-Healing MLOps Platform for Fraud Detection

[GitHub](#)

Python · Scikit-learn · MLflow · SHAP · Docker

- Architected a real-time, containerized fraud-detection service with automated model retraining and **MLflow**-based governance for self-healing pipeline behavior.
- Integrated a **SHAP**-based explainability layer to expose model decision logic for transparent, auditable predictions.

Production-Grade NLP Text Classification Pipeline (MLOps)

[GitHub](#)

Python · PyTorch · FastAPI · MLflow · Docker · GitHub Actions

- Built end-to-end pipeline (ingestion → training → serving → monitoring) with automated retraining, cutting iteration cycles by **50%**.
- Served models via **FastAPI** and automated deployment with **GitHub Actions** CI/CD, reducing release steps from **8 to 2**.

RailTrack Pro — Industrial Rail Monitoring Platform (SIH 2025)

[GitHub](#) | [Certificate](#)

REST APIs · Data Pipelines · Docker · CI/CD

- Built production-grade industrial monitoring platform with scalable **data pipelines** and **REST APIs** for real-world railway infrastructure use cases.
- Cleared internal Smart India Hackathon rounds; presented system design and architecture to a panel of industry evaluators, collaborating closely with a cross-functional team under time constraints.

Achievements & Highlights

- **Competitive Programming:** Solved **200+** DSA problems on LeetCode — strong grounding in algorithmic problem-solving, time/space complexity analysis, and coding-round readiness.
- **Dual Degree:** B.Tech ECE (CGPA **9.1**) alongside IIT Madras B.Sc Data Science — demonstrates ability to manage rigorous, parallel technical coursework.
- **Hackathons:** Smart India Hackathon 2025 finalist round qualifier (RailTrack Pro), collaborating in a team to deliver a working prototype under time constraints.